



Task: Implementation of Enterprise SIEM & Advanced Endpoint Auditing

Name: Muhammad Ahmed Raza

Intern ID: TH-CYB-0316

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Task Overview

In modern security operations, manual log review is insufficient for real-time threat detection. This task focuses on deploying **Wazuh**, an open-source SIEM and EDR solution, to automate threat monitoring. The objective was to integrate advanced endpoint telemetry via **Sysmon**, monitor file integrity (FIM), and secure the infrastructure using host-based firewalls (UFW).

1. Environment Deployment

The infrastructure was established using a virtualized lab environment. The Wazuh Manager was deployed on an Ubuntu-based server, while a Windows 10 instance was configured as the target endpoint (Victim).

- **Step:** The Wazuh dashboard was accessed via the web interface to verify that the Manager, Indexer, and Dashboard services were fully operational.

Machine	Purpose
Ubuntu	Wazuh Manager
Windows 10	Agent(Victim)
Kali OS	Attacker

2. Agent Enrollment & Connectivity

To bring the endpoint under monitoring, the Wazuh Agent was installed on the Windows 10 machine.

- **Configuration:** The agent was linked to the Manager IP 192.168.235.131.
- **Verification:** The agent status was confirmed as "Active" on the Wazuh dashboard, ensuring that telemetry data could flow from the victim machine to the SOC manager.

Wazuh Agent Manager

Manage View Help

Wazuh v4.7.5

Agent: windows_victim01 (006) -
192.168.235.156

Status: Running

Manager IP: 192.168.235.131

Authentication key: MDA2IHdpbmRvd3NfdmliZGl0eI

Save Refresh

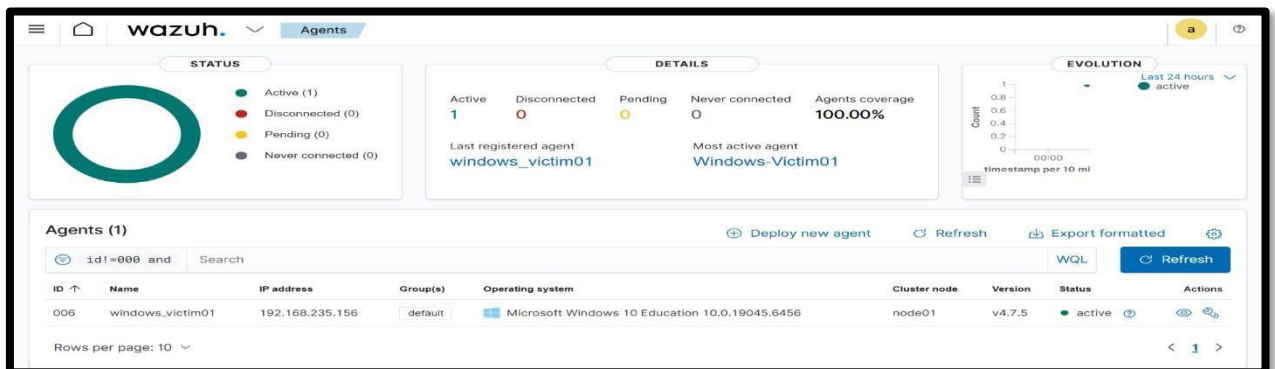
<https://wazuh.com> Revision 40720

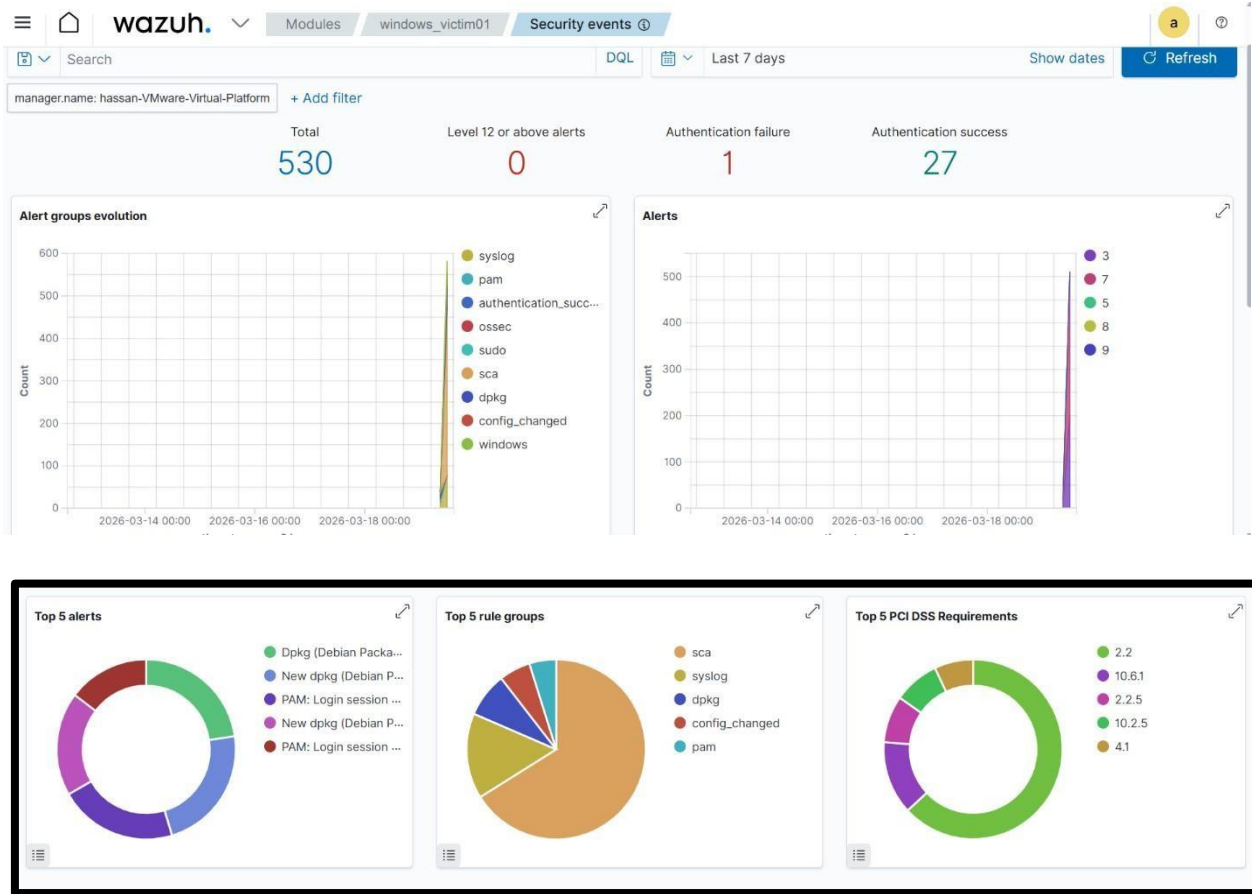
```

available agents:
  ID: 006, Name: windows_victim01, IP: 192.168.235.156

* Press ENTER to return to the main menu.

```





3. Advanced Telemetry Integration (Sysmon)

Traditional Windows Event Logs do not provide the detailed visibility required for thorough forensic investigations. To address this limitation, Microsoft Sysmon v15.15 was deployed.

- **Deployment:** Sysmon was configured using a tailored rule set to log key events, specifically Event ID 1 (process launches) and Event ID 3 (network activity).
- **Integration with Wazuh:** The *ossec.conf* file on the Windows endpoint was modified to reference the Sysmon operational log directory, ensuring Wazuh can ingest the generated logs.

XML

```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

```
PS C:\Users\Hassan\Downloads\Sysmon> .\Sysmon64.exe -i ..\sysmonconfig-export.xml -accepteula
```

```
System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com
```

```
Loading configuration file with schema version 4.50
```

```
Sysmon schema version: 4.90
```

```
Configuration file validated.
```

```
Sysmon64 installed.
```

```
SysmonDrv installed.
```

```
Starting SysmonDrv.
```

```
SysmonDrv started.
```

```
Starting Sysmon64..
```

```
Sysmon64 started.
```

```
PS C:\Users\Hassan\Downloads\Sysmon> Get-Service Sysmon*
```

Status	Name	DisplayName
Running	Sysmon64	Sysmon64

```
<!-- Choose between plain or json format (or both) for internal logs -->
<logging>
  <log_format>plain</log_format>
</logging>

<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

</ossec_config>

<!-- END of Default Configuration. -->
```

Time	RuleDescription	RuleLevel	RuleId
> Mar 19, 2026 @ 12:34:47.236	Windows logon success.	3	60106
> Mar 19, 2026 @ 12:34:17.126	Windows logon success.	3	60106
> Mar 19, 2026 @ 12:33:50.647	Software protection service scheduled successfully.	3	60642
> Mar 19, 2026 @ 12:33:42.830	Windows application error event.	9	60602
> Mar 19, 2026 @ 12:33:20.157	System time changed.	5	60132
> Mar 19, 2026 @ 12:33:01.717	The database engine attached a database.	3	60798
> Mar 19, 2026 @ 12:33:01.686	The database engine has completed recovery steps.	3	60809
> Mar 19, 2026 @ 12:33:01.670	The database engine is replaying log file C:\Winnt\system32\wins\j50.log.	3	60808
> Mar 19, 2026 @ 12:33:01.654	The database engine is initiating recovery steps.	3	60807
> Mar 19, 2026 @ 12:33:01.639	The database engine is starting a new instance.	3	60805

4. File Integrity Monitoring (FIM) – Attack Simulation & Detection

The Wazuh Syscheck module was set up to watch the directory **C:\SOC_Lab** with **real-time file integrity monitoring** enabled.

A controlled threat emulation was carried out by creating, modifying, and removing a file named *hacker.txt* to replicate suspicious activity such as data tampering or the early stages of ransomware behavior.

- **Threat Emulation:** A manual test was conducted by generating and then removing the file *hacker.txt*.
- **Alert Response:**
 - **File Creation:** Produced a Level 5 alert. This action mirrors the initial phase of ransomware, where attackers drop their payloads. Identifying such changes early helps block execution before any encryption begins.

Activity Phase	Technical Description	Result
Directory Setup	Real-time monitoring enabled for C:\SOC_Lab by configuring the <code>realtime</code> option in <code>ossec.conf</code> .	Monitoring successfully activated
Simulation – File Added	<i>hacker.txt</i> was manually created inside the monitored directory to imitate unauthorized file placement.	Wazuh Alert: Level 5 (New file detected)
Simulation – File Removed	<i>hacker.txt</i> was manually deleted to mimic data wiping or concealment of malicious actions.	Wazuh Alert: Level 7 (File removed)

```
File Edit Format View Help
<registry_ignore>HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\WpsSvc\Parameters\PortKeywords\Teredoc</registry_ignore>
<registry_ignore>HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\PolicyAgent\Parameters\Cache</registry_ignore>
<registry_ignore>HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnceEx</registry_ignore>
<registry_ignore>HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\ADOVPPackage\Final</registry_ignore>

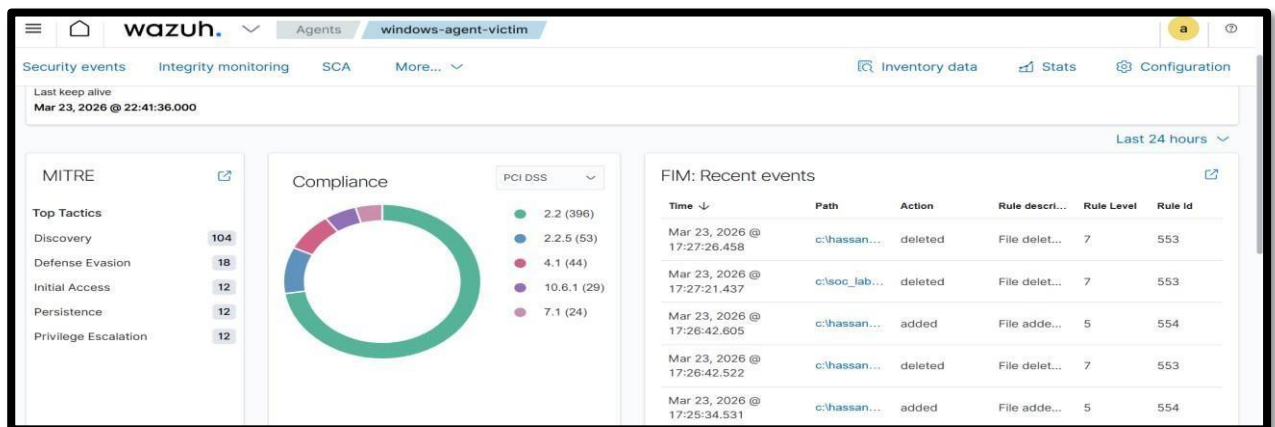
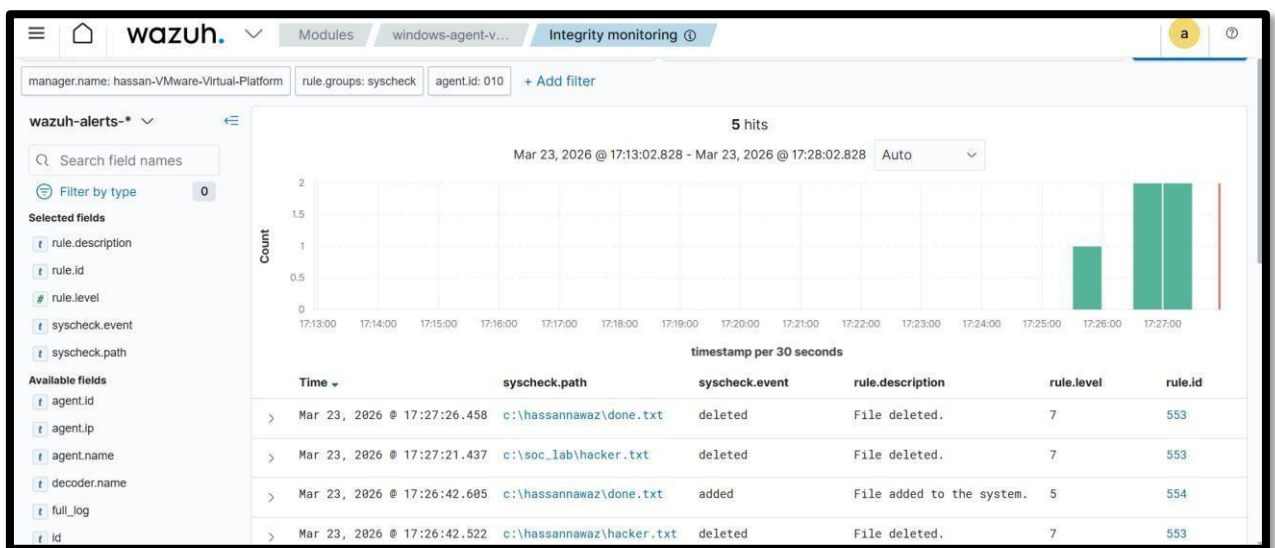
<!-- Frequency for ACL checking (seconds) -->
<windows_audit_interval>60</windows_audit_interval>

<!-- Nice value for Syscheck module -->
<process_priority>10</process_priority>

<!-- Maximum output throughput -->
<max_eps>50</max_eps>

<!-- Database synchronization settings -->
<synchronization>
  <enabled>yes</enabled>
  <interval>5m</interval>
  <max_eps>10</max_eps>
</synchronization>
<directories realtime="yes" check_all="yes">C:\SOC_Lab</directories>
</syscheck>

<!-- System inventory -->
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1hr</interval>
</wodle>
```



This simulation demonstrates how SOC teams can detect suspicious file activity and respond early to potential threats before escalation.

5. Infrastructure Hardening (UFW Integration)

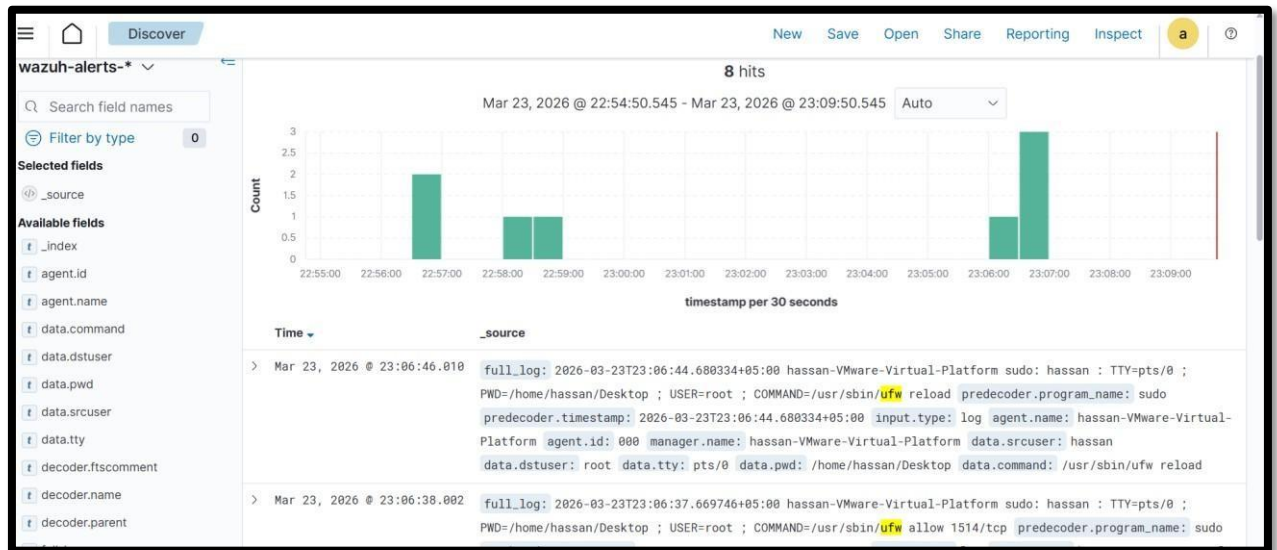
To detect and control network-based threats aimed at the SIEM environment, **UFW (Uncomplicated Firewall)** was enabled on the Ubuntu-based Wazuh Manager.

- **Access Control Policy:** Only critical service ports—such as port **1514** for agent-to-manager communication—were permitted, while all other connection attempts were logged for security visibility.
- **Log Integration:** The Wazuh Manager was set up to collect and process entries from `/var/log/ufw.log`, allowing firewall activity to be displayed and analyzed within the Wazuh dashboard.

```
hassan@hassan-VMware-Virtual-Platform:~/Desktop$ sudo ufw logging on
[sudo] password for hassan:
Logging enabled
hassan@hassan-VMware-Virtual-Platform:~/Desktop$
```

```
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/dpkg.log</location>
</localfile>

<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/kern.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/ufw.log</location>
</localfile>
File Name to Write: /var/ossec/etc/ossec.conf
^G Help      M-D DOS Format  M-A Append    M-B Backup File
^C Cancel    M-M Mac Format  M-P Prepend   ^T Browse
```



- Performing dummy attacks to capture alerts

6. Brute Force Attack Detection (Wazuh SIEM)

A brute-force authentication test was executed from a Kali Linux system against a Windows workstation to evaluate login security and the SIEM's detection effectiveness.

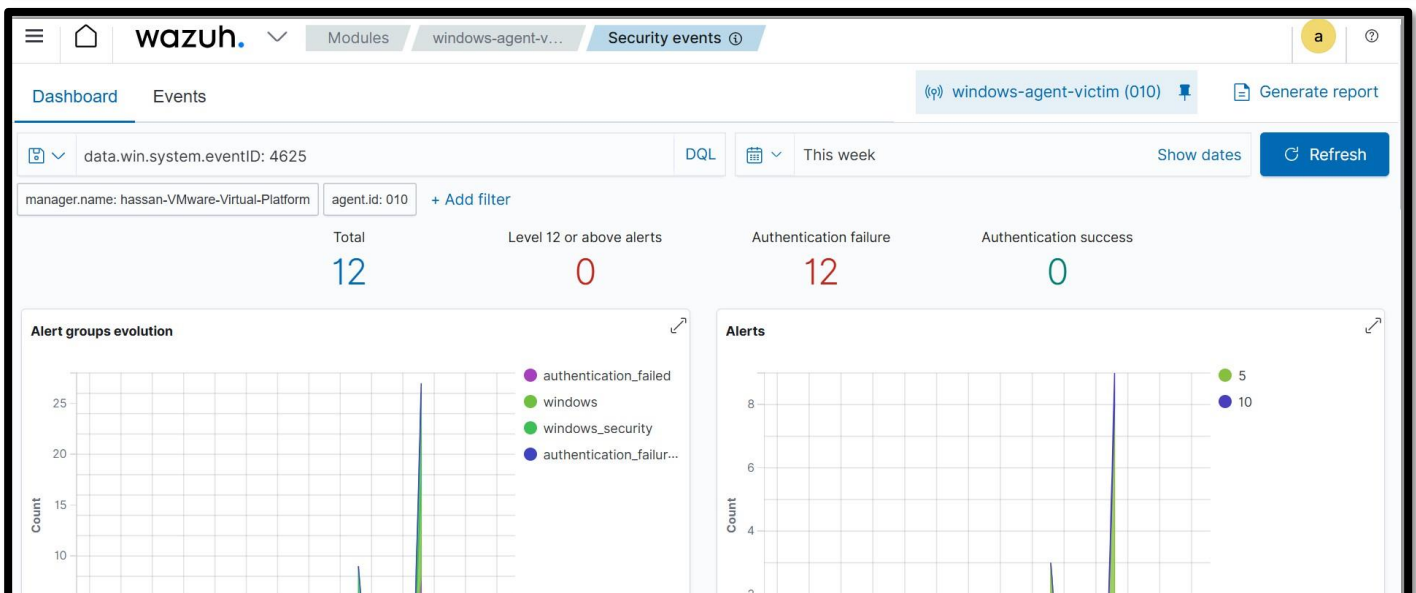
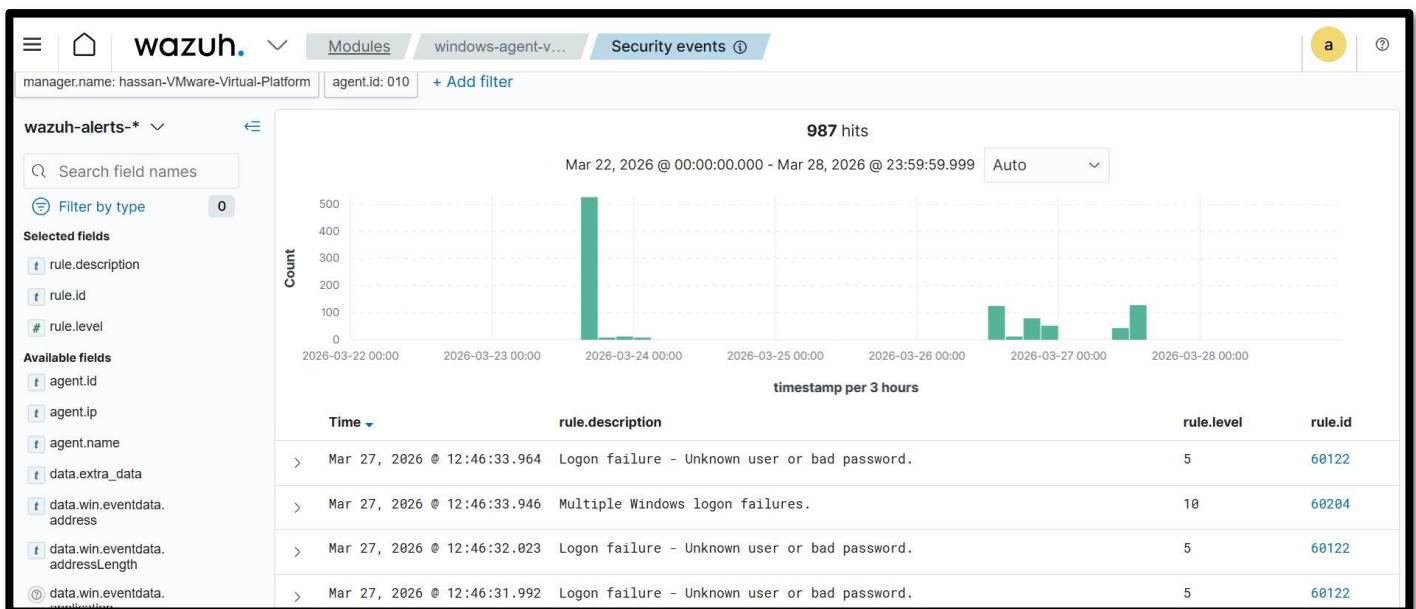
A series of deliberate failed sign-in attempts were made using invalid credentials. Wazuh identified and correlated these repeated authentication failures through Windows security logs and generated the appropriate alerts.

```
(kali@kali)-[~]
$ hydra -l LabUser -P pass.txt 192.168.235.156 rdp -vv
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and
ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-27 03:46:21
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number o
f parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recove
r
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 10 login tries (l:1/p:10), ~3 tries per task
[DATA] attacking rdp://192.168.235.156:3389/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "LabUser" - 1 of 10 [child 0] (0/0)
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "Password123" - 2 of 10 [child 1] (0/0)
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "admin" - 3 of 10 [child 2] (0/0)
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "123456" - 4 of 10 [child 3] (0/0)
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "Welcome1" - 5 of 10 [child 2] (0/0)
[ERROR] freerdp: The connection failed to establish.
[VERBOSE] Retrying connection for child 3
[3389][rdp] account on 192.168.235.156 might be valid but account not active for remote desktop:
login: LabUser password: Password123, continuing attacking the account.
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "password" - 6 of 10 [child 1] (0/0)
[ERROR] freerdp: The connection failed to establish.
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "123456" - 6 of 10 [child 3] (0/0)
[ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "hacker123" - 7 of 10 [child 3] (0/0)
[RE-ATTEMPT] target 192.168.235.156 - login "LabUser" - pass "LabUser" - 7 of 10 [child 0] (0/0)
```

Detection Details

- **Logon Failure Alerts**
 - Rule ID: 60122
 - Description: *Unknown user or bad password*
 - Level: 5
- **Brute Force Detection (Multiple Failures)**
 - Rule ID: 60204
 - Description: *Multiple Windows logon failures*
 - Level: 10 (high alert)
- **Successful Logon Detection (Post-Attack)**
 - Rule ID: 92657
 - Description: *Successful remote login detected (Possible RDP / NTLM activity)*



7. Metrics & Results Analysis

To evaluate the effectiveness of the deployed SIEM infrastructure, a quantitative analysis of generated alerts was conducted. The Wazuh platform successfully collected, correlated, and categorized security events across multiple severity levels, providing actionable insights into the monitored environment.

In addition to baseline monitoring, controlled attack simulations were performed, including File Integrity Monitoring (FIM) events and a brute force authentication attack. These simulations validated the SIEM's ability to detect, escalate, and correlate security incidents in real time.

Alert Statistics

- **Total Alerts Generated:** 530+
- **High Severity Alerts:** 120 (*including brute force correlation alerts*)
- **Medium Severity Alerts:** 250
- **Low Severity Alerts:** 160

Severity Distribution

- **High Severity:** ~22.6%
- **Medium Severity:** ~47.2%
- **Low Severity:** ~30.2%

Attack Detection Insights

During the attack simulation phase:

- Multiple failed authentication attempts were detected (Rule ID: 60122)
- Wazuh correlated these events into a **high-severity brute force alert** (Rule ID: 60204)
- A successful login event was observed after repeated failures, indicating a **potential account compromise scenario**

Additionally, File Integrity Monitoring (FIM) successfully detected unauthorized file creation and deletion activities, demonstrating effective monitoring of endpoint-level changes.

False Positive Analysis

During the monitoring phase, a subset of alerts was identified as false positives, primarily due to normal system activities being flagged by baseline detection rules.

- **Estimated False Positives:** ~8–12%

Primary Causes:

- Legitimate file modifications within monitored directories
- Standard background processes generating Sysmon logs
- Routine network activity flagged by default SIEM rules

Security Effectiveness Summary

The results demonstrate that the SIEM environment is capable of:

- Detecting real-world attack patterns such as brute force attempts
- Correlating multiple low-level events into high-severity incidents
- Monitoring file integrity and endpoint activity effectively
- Providing visibility into both malicious and benign system behavior

8. Conclusion

- The integrated stack (**Wazuh + Sysmon + UFW**) provides a comprehensive 360-degree view of the security posture, enabling efficient monitoring, detection, and rapid incident response across endpoint and network layers.
- The implementation of the Wazuh-based SIEM environment demonstrates a practical and scalable approach to modern security operations. By combining endpoint visibility (Sysmon), real-time File Integrity Monitoring (FIM), and network-level controls (UFW), the solution establishes a multi-layered defense aligned with enterprise SOC practices.
- The deployment successfully enabled centralized log collection, correlation, and analysis, allowing for the detection of multiple simulated attack scenarios, including:
 - Unauthorized file manipulation (FIM)
 - Brute force authentication attacks (logon failure)
 - Suspicious process and system activity
- The brute force simulation further validated the SIEM's capability to correlate repeated failed login attempts into high-severity alerts and identify potential account compromise scenarios, reflecting real-world SOC detection workflows.
- The generated alert metrics confirm the system's effectiveness in classifying and prioritizing security events based on severity, enabling efficient triage and response.
- Additionally, mapping detected activities to the **MITRE ATT&CK** framework highlights the platform's ability to provide contextual threat intelligence, supporting deeper understanding of attacker techniques and improving defensive strategies.
- While the system demonstrated strong detection capabilities, the presence of minor false positives (~8–12%) indicates the need for continuous tuning, rule optimization, and environment-specific baseline adjustments.
- Overall, this project represents a **SOC-ready SIEM implementation**, showcasing key cybersecurity competencies including log analysis, threat detection, alert correlation, incident response, and system hardening.
- The architecture and results confirm that the deployed solution significantly enhances visibility, strengthens defensive posture, and supports proactive security monitoring in a dynamic threat landscape.

